Take Home Assignment

Problem Statement

You will be provided with a collection of conveyor belt images captured at different times of the day. The objective is to develop a model that can automatically identify and flag instances of belt damage.

The training dataset consists of images captured at different times of the day and includes both undamaged and damaged conveyor belt instances.

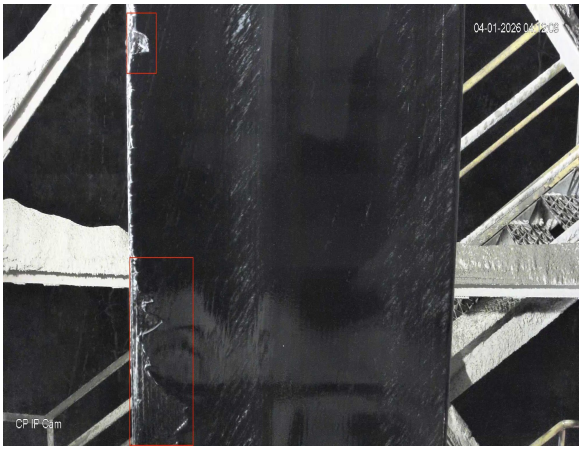
The belt may contain scratches of an undefined nature; therefore, the scope of this assignment is limited to detecting **scratch** and **edge_damage** defects only.

Dataset

- The training dataset is provided in YOLO format.
- It includes image files along with corresponding .txt label files.
- The label files contain annotations describing the outline of the conveyor belt.
- You may use these provided annotations to extract or utilise the belt region as part of your approach.

Sample Images

The images below are representative samples of conveyor belt damage from the dataset:



Night-time capture — scratch damage (happens on the surface)

Daytime capture — edge damage (always on the edge of the belt)

Inference Pipeline

Deliver a Python script (pipeline.py) with the following interface:

```
python pipeline.py --image_dir <path_to_image_folder> --output_dir <folder>
```

Expected Output

For each input image, the pipeline must produce the following:

Output	Description
`Annotated Image`	Original image with bounding boxes overlaid on each detected damage region, saved as <original_image_name>.jpg.
`detections.json`	One JSON file per image, saved as <original_image_name>.json. Format:

```
{
  "1": {
    "bbox_coordinates": [x_min, y_min, x_max, y_max]
  },
  "2": { ... }
}
```

bbox_coordinates: [x_min, y_min, x_max, y_max] in pixels. The detection index key holds no significance beyond identification.

Accuracy Metric

Model performance will be evaluated using the **Mean F1 Score** computed across a range of Intersection over Union (IoU) thresholds, denoted as **mF1@0.5–0.95**. The evaluation procedure is as follows:

- **Bounding Box Matching** — Each predicted bounding box is matched to its corresponding ground-truth bounding box using a greedy assignment strategy based on maximum IoU overlap.
- **IoU-Based Classification** — For each matched pair, the Intersection over Union (IoU) is computed. A prediction is classified as a True Positive (TP) if its IoU meets or exceeds the given threshold; otherwise, it is classified as a False Positive (FP). Unmatched ground-truth boxes are counted as False Negatives (FN).
- **Threshold Sweep** — The classification step is repeated across IoU thresholds ranging from 0.50 to 0.95, in increments of 0.05.
- **F1 Computation** — At each threshold, the F1 Score is calculated as the harmonic mean of Precision and Recall.
- **Final Metric** — The final evaluation metric, **mF1@0.5–0.95**, is obtained by averaging the F1 Scores across all IoU thresholds. This provides a robust and comprehensive measure of detection accuracy at varying levels of localization strictness.

Deliverables

Item	Description
`pipeline.py`	Inference script — clean, modular, runnable from CLI
`train.py / notebook`	Training script with instructions to reproduce
`model_weights`	Trained model weights (shareable link if large)
`outputs/`	Annotated images + detections.json for all dataset images
`README.md`	Setup, training, and inference reproduction steps